Technical University of Denmark

Written examination date: May 23, 2023

Course title: Computer Science Modelling

Course number: 02141

Aids allowed: All aid

Exam duration: 4 hours

Weighting: 7-step scale

NOTES:

- You can submit your answers digitally or as a paper submission.
- Digital submissions must be done as a single pdf-file.
- You must clearly indicate on each page your student number, and which task you attempt to solve.
- We recommend that you first skim all tasks before deciding in which order you want to solve them.

Answers Included

Exercise 1 (15%) Semantics

Consider the program graphs `PG1` - `PG3` in Figures 1 - 3 on page 3.

Question 1a: *For each program graph* `PGi` *(*`i` $\in \{1, 2, 3\}$*) on page 3, determine whether there exists a program C_i in Guarded Command Language such that* **edges**$(q_{\triangleright} \rightsquigarrow q_{\blacktriangleleft})[\![C_i]\!]$ *generates it. If a suitable program exists, provide it. Otherwise, argue (in 1-2 sentences) why no such program exists.*

Solution : Figure 1: No such program exists because the edge from the final node back to the initial node is not $\neg(x > 0)$.

Figure 2:

```
y := 23;
do
  x > 0 -> x := x + 1
  [] x < 0 -> x := x - 1
  [] x > 0 -> y := 17; x := x - 1
od;
if
  y = 17 -> z := 13
  [] x != 0 -> z := 7
fi
```

Figure 3: No such program exists because the edge into the final node is a guard that is not followed by a command.

Question 1b: *For each of the program graphs* `PG1` - `PG3` *in Figures 1 - 3 on page 3, determine whether it is a deterministic system.*

Justify your answer in one sentence.

Solution : Figure 1: deterministic because every node has at most one outgoing edge.

Figure 2: deterministic. The overlapping criterion (proposition 1.22) does not help us. For example, q_2 has two outgoing edges and the semantics of their actions is defined for any memory with $y = 17$ and $x \neq 0$. However, if we look closely at the definition and consider only complete execution sequences (def. 1.21), then this is not an issue. Indeed, the PG has no complete execution sequence and is therefore deterministic.

Figure 3: deterministic. Only the initial node has multiple outgoing edges. The semantics of their actions do not overlap.

Question 1c: *Consider the program graph* `PG2` *in Figure 2 on page 3. How many different execution sequences of the form*

$$\langle q_{\triangleright}; \sigma \rangle \stackrel{\omega}{\Longrightarrow}^{*} \langle q_{\blacktriangleleft}; \sigma' \rangle$$

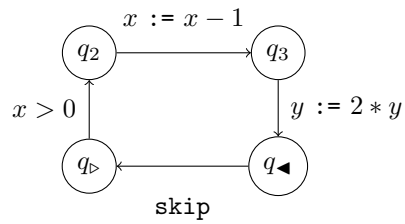


Figure 1: Program graph PG1

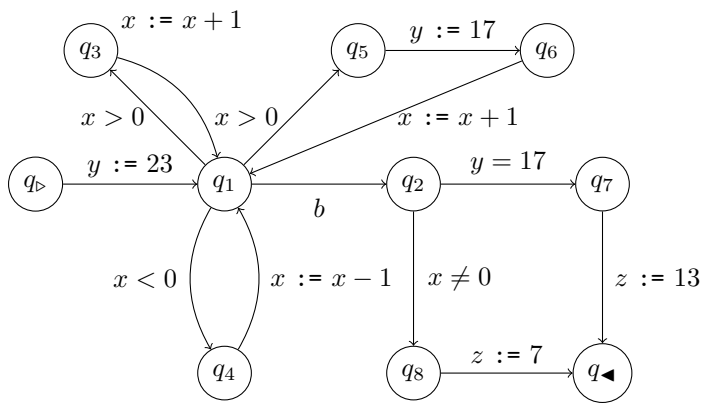


Figure 2: Program graph PG2 with $b = \neg(x > 0) \wedge \neg(x < 0) \wedge \neg(x > 0)$.

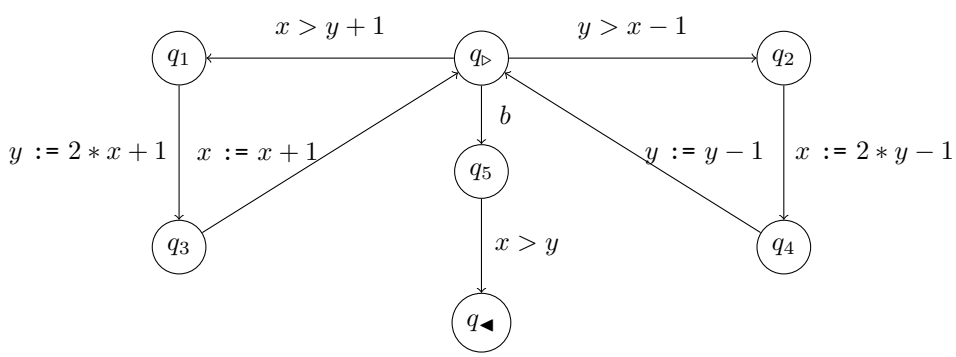


Figure 3: Program graph PG3 with $b = \neg(x > y + 1) \wedge \neg(y > x - 1)$.

exist if the set of variables is $\mathbf{Var} = \{x, y, z\}$*?*

Justify your answer in 1-3 sentences.

Solution :

There is no complete execution sequence. To reach q_2, we need $x = 0$. Then, we cannot reach q_8 and thus also need $y = 17$. In q_7, z is assigned 13. Hence, every final memory must be of the form $(0, 17, 13)$. y is initially set to 23 and can only be set to 17 by reaching q_5 if $x > 0$, which can then not be reset to 00.

Question 1d: *Prove or disprove (in 1-5 lines): every program graph that constitutes a deterministic and evolving system has at least one complete execution sequence.*

Solution :

This is not correct. For example, the program graph obtained from

```
do
  true - > skip
od
```

is deterministic and evolving but does not have any complete execution sequences.

Question 1e: *Suppose we extend the Guarded Command Language by a new command for Java-style for loops*

$$\texttt{for}(i \texttt{ := } a_1;\ b;\ i \texttt{ := } a_2)\ \{\ C\ \}$$

where i is an (integer) variable, a_1, a_2 are arithmetic expressions, b is a Boolean expression, and C is a command. Informally, the above loop

1. *sets variable i to the value of arithmetic expression a_1;*
2. *executes the loop body C if the loop guard b holds (otherwise, the loop terminates);*
3. *updates i to the value of a_2 after termination of loop body C; and*
4. *continues with step (2).*

Formalise the above informal semantics by defining the program graph of for-loops. That is, give a formal definition of

$$\mathbf{edges}(q_{\triangleright} \rightsquigarrow q_{\blacktriangleleft})[\![\texttt{for}(i \texttt{ := } a_1;\ b;\ i \texttt{ := } a_2)\ \{\ C\ \}]\!].$$

Briefly explain your definition in 1-3 sentences.

Solution :

$$
\begin{aligned}
& \mathbf{edges}(q_{\triangleright} \rightsquigarrow q_{\blacktriangleleft})[\![\texttt{for}(i \texttt{ := } a_1;\ b;\ i \texttt{ := } a_2)\ \{\ C\ \}]\!] \\
= \ & \mathbf{edges}(q_{\triangleright} \rightsquigarrow q)[\![i \texttt{ := } a_1]\!] \\
& \quad \cup\ \{(q, b, q')\} \\
& \quad \cup\ \{(q, \neg b, q_{\blacktriangleleft})\} \\
& \quad \cup\ \mathbf{edges}(q' \rightsquigarrow q'')[\![\ C\ ]\!] \\
& \quad \cup\ \mathbf{edges}(q'' \rightsquigarrow q)[\![i \texttt{ := } a_2]\!],
\end{aligned}
$$

where q, q', q'' are fresh nodes. The solution defines the program graph as in the four steps of the task. It can, for example, be obtained from rewriting the for-loop as

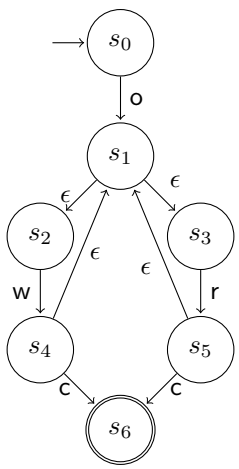
```
i := a1; do b -> C; i := a2 od
```

and then computing its program graph.

Exercise 2 (20%) Formal Languages

Alice and Bob want to formalise correct usages of an API with endpoints o (open), w (write), r (read), and c (close).

Alice proposes automaton A:



Bob proposes regular expression B:

$\mathsf{or}^*\mathsf{w}^*\mathsf{c}$

where the set $\{\mathsf{o}, \mathsf{r}, \mathsf{w}, \mathsf{c}\}$ is the input alphabet for A and B.

(a) Fill the table below. In the empty cell corresponding to language L provide a short as possible word that belongs to L. Write "———" if no such word exists.

Language	Shortest word
$\mathcal{L}(\mathsf{A}) \cup \mathcal{L}(\mathsf{B})$	
$\mathcal{L}(\mathsf{A}) \cap \mathcal{L}(\mathsf{B})$	
$\mathcal{L}(\mathsf{A}) \setminus \mathcal{L}(\mathsf{B})$	
$\mathcal{L}(\mathsf{B}) \setminus \mathcal{L}(\mathsf{A})$	
$\overline{\mathcal{L}(\mathsf{A})} \cap \overline{\mathcal{L}(\mathsf{B})}$	
$\mathcal{L}(\mathsf{A})^* \cup \mathcal{L}(\mathsf{B})^*$	

Solution:

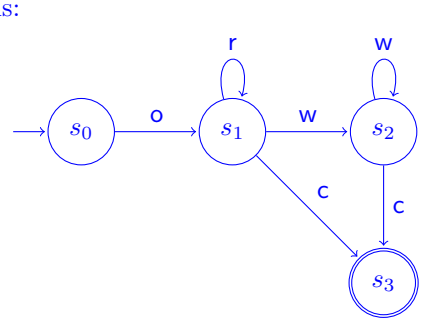
Language	Shortest word
$\mathcal{L}(\mathsf{A}) \cup \mathcal{L}(\mathsf{B})$	oc
$\mathcal{L}(\mathsf{A}) \cap \mathcal{L}(\mathsf{B})$	owc
$\mathcal{L}(\mathsf{A}) \setminus \mathcal{L}(\mathsf{B})$	owrc
$\mathcal{L}(\mathsf{B}) \setminus \mathcal{L}(\mathsf{A})$	oc
$\overline{\mathcal{L}(\mathsf{A})} \cap \overline{\mathcal{L}(\mathsf{B})}$	ϵ
$\mathcal{L}(\mathsf{A})^* \cup \mathcal{L}(\mathsf{B})^*$	ϵ

(b) Alice and Bob decide to transform the regular expression B into a finite state automaton to easily compare their solutions. Alice correctly applies the transformations seen in class to obtain a minimized DFA $\mathsf{A_B}$.

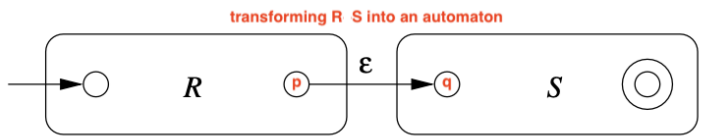
Provide $\mathsf{A_B}$ as transition diagram. Do *not* include the so-called "dead" / "error" state and its transitions. Explain at a high-level which transformations you have applied (1 sentence per transformation). You do not need to provide the details of each transformation.

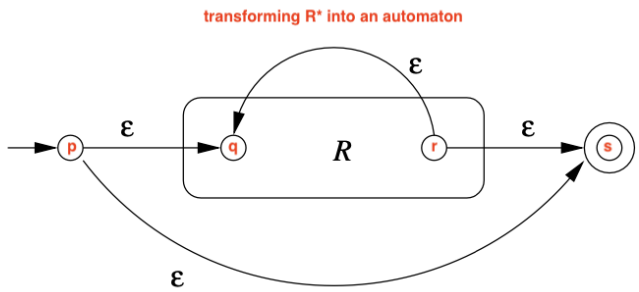
Solution:

The solution is obtained by applying the following transformations: regular expression into ϵ-NFA, ϵ-NFA into DFA (subset construction), and minimization of DFA. The final result (modulo renaming of the state names) is:



(c) Bob does not like to have too many ϵ transitions and decides to modify the algorithm for transforming regular expressions into finite automata by changing the rules below:



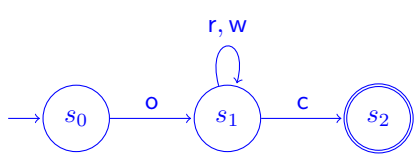


The modifications of Bob are:

- In the transformation of RS, states p and q are merged.
- In the transformation of R^*, p is merged with q, and r is merged with s.

After applying the new algorithm, Bob obtains the automaton $\mathsf{B_B}$. Provide $\mathsf{B_B}$ as minimized DFA (again as a transition diagram, with no "error"/"dead" state). You do not need to explain how you obtained it.

Solution:



(d) Answer the following questions with a short sentence (as a rule of thumb, your answer should fit in the table). The sentence must start with "YES" or "NO".

Question	Short answer
$\mathcal{L}(\mathsf{A_B}) = \mathcal{L}(\mathsf{B_B})$?	
$\mathcal{L}(\mathsf{A_B}) = \mathcal{L}(\mathsf{B})$?	
$\mathcal{L}(\mathsf{B_B}) = \mathcal{L}(\mathsf{B})$?	
$\mathcal{L}(\mathsf{A_B}) = \mathcal{L}(\mathsf{A})$?	
$\mathcal{L}(\mathsf{B_B}) = \mathcal{L}(\mathsf{A})$?	

Solution:

Question	Short answer
$\mathcal{L}(\mathsf{A_B}) = \mathcal{L}(\mathsf{B_B})$?	No, since owrc is in $\mathcal{L}(\mathsf{B_B})$ but not in $\mathcal{L}(\mathsf{A_B})$
$\mathcal{L}(\mathsf{A_B}) = \mathcal{L}(\mathsf{B})$?	Yes, since Alice applied correct transformations
$\mathcal{L}(\mathsf{B_B}) = \mathcal{L}(\mathsf{B})$?	No (follows from the two facts above)
$\mathcal{L}(\mathsf{A_B}) = \mathcal{L}(\mathsf{A})$?	No (follows from the first fact and the answers to (a))
$\mathcal{L}(\mathsf{B_B}) = \mathcal{L}(\mathsf{A})$?	No, since oc is in $\mathcal{L}(\mathsf{B_B})$ but not in $\mathcal{L}(\mathsf{A})$

Exercise 3 (20%) Program Verification

Question 3a: *Complete the following annotated GCL program to a formal program proof (that is, all annotations must result from applying the proof rules of Floyd-Hoare logic introduced in the course).*

```
{ n ≥ 0 }
r := 12 * n;
k := 0;
do { k ≤ 6 * n ∧ r = 12 * n − 2 * k }
        k < 6 * n ->
            r := r − 2;
            k := k + 1
    []
        k < 6 * n ->
            k := k + 3;
            k := k − 3
od
{ r = 0 }
```

Solution :

$\{\ n \geq 0\ \}$

r `:=` $12 * n$;

$\{\ \exists \underline{r}\colon n \geq 0 \wedge r = 12 * n\ \}$

$\{\ n \geq 0 \wedge r = 12 * n\ \}$

k `:=` 0;

$\{\ \exists \underline{k}\colon n \geq 0 \wedge r = 12 * n \wedge k = 0\ \}$

$\{\ n \geq 0 \wedge r = 12 * n \wedge k = 0\ \}$

$\{\ k \leq 6 * n \wedge r = 12 * n - 2 * k\ \}$

`do` $\{\ k \leq 6 * n \wedge r = 12 * n - 2 * k\ \}$

$\{\ k \leq 6 * n \wedge r = 12 * n - 2 * k\ \}$

$k < 6 * n$ `->` $\{\ k < 6 * n \wedge k \leq 6 * n \wedge r = 12 * n - 2 * k\ \}$

$\{\ k < 6 * n \wedge r = 12 * n - 2 * k\ \}$

r `:=` $r - 2$;

$\{\ \exists \underline{r}\colon k < 6 * n \wedge \underline{r} = 12 * n - 2 * k \wedge r = \underline{r} - 2\ \}$

k `:=` $k + 1$

$\{\ \exists \underline{k}\colon \exists \underline{r}\colon \underline{k} < 6 * n \wedge \underline{r} = 12 * n - 2 * \underline{k} \wedge r = \underline{r} - 2 \wedge k = \underline{k} + 1\ \}$

$\{\ (k - 1) < 6 * n \wedge (r + 2) = 12 * n - 2 * (k - 1)\ \}$

$\{\ k \leq 6 * n \wedge r = 12 * n - 2 * k\ \}$

`[]`

$k < 6 * n$ `->` $\{\ k < 6 * n \wedge k \leq 6 * n \wedge r = 12 * n - 2 * k\ \}$

$\{\ k < 6 * n \wedge r = 12 * n - 2 * k\ \}$

k `:=` $k + 3$;

$\{\ \exists \underline{k}\colon \underline{k} < 6 * n \wedge r = 12 * n - 2 * \underline{k} \wedge k = \underline{k} + 3\ \}$

$\{\ (k - 3) < 6 * n \wedge r = 12 * n - 2 * (k - 3)\ \}$

k `:=` $k - 3$

$\{\ \exists \underline{k}\colon (\underline{k} - 3) < 6 * n \wedge r = 12 * n - 2 * (\underline{k} - 3) \wedge k = \underline{k} - 3\ \}$

$\{\ k \leq 6 * n \wedge r = 12 * n - 2 * k\ \}$

$\{\ k \leq 6 * n \wedge r = 12 * n - 2 * k\ \}$

`od`

$\{\ k \leq 6 * n \wedge r = 12 * n - 2 * k \wedge \neg(k < 6 * n) \wedge \neg(k < 6 * n)\ \}$

$\{\ r = 0\ \}$

Question 3b: *Show that the GCL program below computes n^3 by constructing a formal program proof for the given contract (that is, all annotations must result from applying the proof rules of Floyd-Hoare logic introduced in the course).*

$\{$ `true` $\}$
p := $n * n$;
r := 0;
`do`
  $p > 0$ `->`
    r := $r + n$;
    p := $p - 1$
`od`
$\{ r = n^3 \}$

Solution :

$\{$ `true` $\}$
p := $n * n$;
$\{ \exists \underline{p}\colon \texttt{true} \wedge p = n * n \}$
$\{ p = n * n \}$
r := 0;
$\{ \exists \underline{r}\colon p = n * n \wedge r = 0 \}$
$\{ r = n^3 - p * n \wedge p \geq 0 \}$
`do` $\{ r = n^3 - p * n \wedge p \geq 0 \}$
  $p > 0$ `->` $\{ r = n^3 - p * n \wedge p \geq 0 \wedge p > 0 \}$
    $\{ r = n^3 - p * n \wedge p > 0 \}$
    r := $r + n$;
    $\{ \exists \underline{r}\colon \underline{r} = n^3 - p * n \wedge p > 0 \wedge r = \underline{r} + n \}$
    $\{ (r - n) = n^3 - p * n \wedge p > 0 \}$
    p := $p - 1$
    $\{ \exists \underline{p}\colon (r - n) = n^3 - \underline{p} * n \wedge \underline{p} > 0 \wedge p = \underline{p} - 1 \}$
    $\{ (r - n) = n^3 - (p + 1) * n \wedge (p + 1) > 0 \}$
    $\{ r = n^3 - p * n \wedge p \geq 0 \}$
  $\{ r = n^3 - p * n \wedge p \geq 0 \}$
`od`
$\{ r = n^3 - p * n \wedge p \geq 0 \wedge \neg(p > 0) \}$
$\{ r = n^3 \}$

Question 3c: *The course material does not consider a proof rule for verifying assignments of the form $A[x]$:= a, where A is an array, x is a variable and a is an arithmetic expression. Suppose we treat an assignment to an array element like any other assignment, that is, we construct the triple*

$$\{\ P\ \}A[x]\ :=\ a\{\ \exists \underline{y}\colon P[\underline{y}\,/\,A[x]]\ \wedge\ A[x] = a[\underline{y}\,/\,A[x]]\ \}\ ,$$

where $P[\underline{y}\,/\,A[x]]$ and $a[\underline{y}\,/\,A[x]]$ denote the substitution (i.e. syntactic replacement) of every occurrence of $A[x]$ by $\underline{y}$ in P and a, respectively.

*Show that the above rule is **not** correct. To this end, find a GCL program C, predicates P and Q, and memories σ and σ' such that*

- *one can construct a program proof $\{\ P\ \}\ C\ \{\ Q\ \}$ using our existing rules and the above rule for assigning to array elements, and*
- *there is a complete execution sequence of C with initial memory σ and final memory σ' such that $\sigma \models P$ and $\sigma' \not\models Q$.*

Give both the program proof and the complete execution sequence for your example to demonstrate that you have indeed chosen adequate C, P, Q, σ, σ'.

Hint: Notice that $A[i]$ and $A[j]$ are syntactically different array elements but refer to the same array element if variables i and j store the same value.

Solution :

According to the above rule, we can construct the following triple:

$$\begin{array}{l}\{\ A[1] = 17 \wedge x = 1\ \}\\ A[x]\ :=\ 42\\ \{\ \exists \underline{y}\colon A[1] = 17 \wedge x = 1 \wedge A[x] = 42\ \}\end{array}$$

The postcondition is clearly contradictory because $A[1]$ cannot be both 17 and 42. For a concrete example execution let σ be given by $\sigma(A[1]) = 17$, $\sigma(x) = 1$. Then $\sigma \models A[1] = 17 \wedge x = 1$, $\sigma' = \sigma[A[1] \mapsto 42]$, and

$$\sigma' \not\models \exists \underline{y}\colon A[1] = 17 \wedge x = 1 \wedge A[x] = 42.$$

Exercise 4 (10%) Program Analysis

Question 4a: *Explain whether the following statements about the program graphs* `PG4` *and* `PG5` *in Figures 5 - 7 (on page 15) are* `true`, `false`, *or* `unknown` *(that is, no conclusive answer is possible). Your answers must be* based on the results of the detection of signs analysis *in [FM, chapter 4]. You can safely assume that formalmethods.dk implements this analysis. Explain your answer to each question in 1-3 sentences.*

(a) `PG4` has no complete execution sequences.

(b) For every execution sequence of `PG4`, the sign of `y` will eventually become equal to the sign of `z`.

(c) For every complete execution sequence of `PG5`, no element of array A is 0 whenever x is positive upon termination.

(d) There exist executions of `PG5` that can get stuck.

Solution :

(a) True. A detection of signs analysis for all (27) possible initial memories does not assign any abstract memory to the final node.

(b) Unknown. The detection of signs analysis also admits cases in all reachable nodes in which y and z have different signs. We do not know whether those cases are indeed reachable.

(c) Unknown. The detection of signs analysis admits cases in the final node in which x is positive and A has $\{0, -, +\}$ as possible signs. We do not know if such cases are indeed reachable.

(d) True. If x is initially negative, then execution gets stuck and no signs are assigned to any nodes except the initial one.

Question 4b: *Consider the abstract memory* $\hat{\sigma}[x \mapsto +][y \mapsto +]$ *and the following guarded command:*

$$\texttt{if } x > y \texttt{ -> } x \texttt{ := } x - y \ [\,]\ x \leq y \texttt{ -> } x \texttt{ := } 17 \texttt{ fi}$$

What are the possible signs of x *upon termination according to a detection of signs analysis (as defined in [FM, chapter 4]) for the above initial abstract memory? For each of those signs, explain in 1-3 sentences whether it is realistic (that is if there exists a concrete execution sequence in which we can observe variables whose values have those signs).*

Solution :

The possible signs are $-$, 0, and $+$.

```
do x<0 -> y:=(-1*z)*z
[] x=0 -> y:=0
[] x>0 -> y:=z*z
od
```

Figure 4: Code for PG4

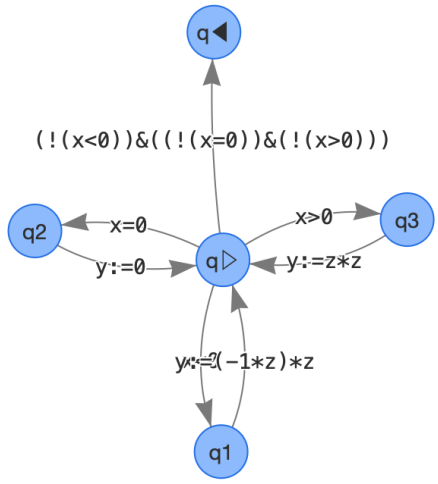


Figure 5: Program graph PG4

```
if
   x >= 0 -> A[x] := A[x+1]
[] x > 1 -> A[x+1] := A[x]-1
fi;
if
   A[x] > 0 -> x := 0
[] A[x] < 0 -> x := 1
fi
```

Figure 6: Code for PG5

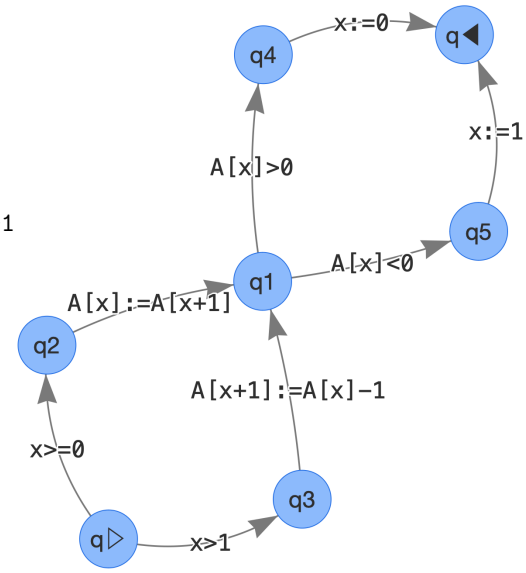


Figure 7: Program graph PG5

$-$ and 0 are not realistic while $+$ is: If we take the first guarded branch, then $x > y$ holds and thus $x := x - y$ should assign a positive number to x. Otherwise, x is always assigned a positive number.

Exercise 5 (10%) Language-based Security

In this exercise we focus on the classical confidentiality policy with public and private variables. Let `h` be a private variable and `l` be a public variable. Alice does not like that the program

```
l := h ; l := 0
```

is considered as *not secure* with the security analysis seen in the course, while it is considered as *non-interferent*. To remedy this, she proposes a new security analysis based on the following (underlined) modification to the function actualFlows (see essential exercises in the lecture on language-based security):

$$\begin{aligned}&\mathsf{actualFlows}(c_1;c_2, X)\\ &\quad = (\mathsf{actualFlows}(c_1, X)\underline{\setminus\{\{y\}\rightrightarrows\mathsf{updates}(c_2) \mid y \in V\}}) \cup \mathsf{actualFlows}(c_2, X)\end{aligned}$$

where V is the set of all variables, and function updates is intended to determine the set of variables that are *necessarily* updated in a program (i.e. updated in all possible executions):

$$\begin{array}{rcl}\mathsf{updates}(\texttt{skip}) & = & \emptyset\\ \mathsf{updates}(x{:=}e) & = & \{x\}\\ \mathsf{updates}(A[e_1]{:=}e_2) & = & \{A\}\\ \mathsf{updates}(c_1;c_2) & = & \mathsf{updates}(c_1) \cup \mathsf{updates}(c_2)\\ \mathsf{updates}(\texttt{if } gc \texttt{ fi}) & = & \mathsf{updates}(gc)\\ \mathsf{updates}(\texttt{do } gc \texttt{ od}) & = & \emptyset\\ \mathsf{updates}(b \texttt{ -> } c) & = & \mathsf{updates}(c)\\ \mathsf{updates}(gc_1[]gc_2) & = & \mathsf{updates}(gc_1) \cap \mathsf{updates}(gc_2)\end{array}$$

Question 5a: *Would the new analysis determine that the above program* `l := h ; l := 0` *is secure? Write "YES" or "NO" and explain your answer in 1 sentence.*

Solution : Yes. The set of actual flows for the entire program would be the empty set because from the flows of `l := h` we have to remove all flows into `l` since `l` is necessarily updated in the continuation.

Question 5b: *Is there a program that has the non-interference property but that is deemed as not secure by the new analysis function? If the answer is "YES" provide the smallest example you can find and explain in 2-3 lines why it is secure but it does not have the non-interference property. If the answer is "NO", provide your argument in 2-3 lines.*

Solution : Yes. The trivial example is `l := h-h` for the same reasons we have seen in class (the modification of Alice is irrelevant in the example): there is a non-allowed flow `h` $\rightarrow$ `l` but the final value of `l` is 0 for all possible initial memories.

Question 5c: *Bob implements the new security analysis of Alice but does a mistake when computing the* updates *function: He writes* $\cup$ *instead of* $\cap$ *in this case of the function:*

$$\mathsf{updates}(gc_1[]gc_2) \quad = \quad \mathsf{updates}(gc_1) \ \cup \ \mathsf{updates}(gc_2)$$

Provide an example of a program that exploits Bob's bug to deem the program as secure but such that the program does not have the non-interference property. Explain your example in 2-3 lines.

Solution : A simple program (with the same variables as the first one) is:

```
l := h;
if
false -> l:=0
[]
true -> skip
fi
```

Bob's implementation will consider that the `if...fi` continuation *necessarily* updates `l` which actually does not happen in any execution. Hence, the analysis will determine that the set of actual flows is empty, but the program is clearly interferent as the final value of `l` is simply `h`.

Question 5d: *Now, forget about Bob's bug. Are we guaranteed that if a program is deemed as secure with the new definition of Alice, it will be non-interferent? If the answer is "NO" provide the smallest example you can find and explain in 2-3 lines why it is secure but it does not have the non-interference property. If the answer is "YES", provide your argument in 2-3 lines.*

Solution : No. Here is an example, with `l1`, `l2` public and `h` private:

```
l1 := h
l2 := l1
l1 := 0
```

The program is secure. The actual flows is just the singleton $\{\texttt{l1} \rightarrow \texttt{l2}\}$ which is allowed by the confidentiality policy. However, `l2` trivially stores the value of `h` at the end of the program.

Question 5e: *Are the following statements correct or incorrect? Provide your argument in 1 sentence.*

(a) *According to the lecture's security analysis, if* c_1`;`c_2 *is secure then* c_2`;`c_1 *is secure.*

(b) *According to Alice's new security analysis, if* c_1`;`c_2 *is secure then* c_2`;`c_1 *is secure.*

(c) *If* c_1`;`c_2 *is non-interferent then* c_2`;`c_1 *is non-interferent.*

Solution :

(a) Correct. The actual flows of c_1`;`c_2 and c_2`;`c_1 are the same.

(b) Incorrect. The actual flows of `l := h; l:=0` and `l := 0; l:= h` are different.

(c) Incorrect. `l := h; l:=0` is non-interferent but `l := 0; l:=h` is not.

Exercise 6 (15%) Context-free Languages

The following context-free grammar provides a syntax for positive modal logic formulas:

$$
\begin{array}{rcll}
F & \rightarrow & P & \text{(proposition...)} \\
 & | & F \wedge F & \text{(... and ...)} \\
 & | & \Diamond F & \text{(possibly...)} \\
 & | & \Box F & \text{(necessarily ...)} \\
P & \rightarrow & \mathsf{a} \mid \mathsf{b} \mid \mathsf{c} & \text{(some atomic propositions)}
\end{array}
$$

where the set of non-terminal symbols is $N = \{F, P\}$, the set of terminal symbols is $T = \{\Box, \Diamond, \mathsf{a}, \mathsf{b}, \mathsf{c}, \}$ and the initial symbol is F. We call a string accepted by the grammar a "formula".

(a) Design a set of datatypes that are suitable to store abstract representations (ASTs) of formulas and show how the formula

$$\mathsf{a} \wedge \Diamond\Box\mathsf{b}$$

would be represented as a value of your datatype.

Solution: In F# we could define the following types:

```
type F =
       | Proposition of P
       | Not of F
       | And of F * F
       | Possibly of F
       | Necessarily of F
and P =
       | a
       | b
       | c
```

The formula could be represented by

```
And(Proposition(a),Possibly(Not(Necessarily(Proposition(b)))))
```

(b) Consider the following operators for composing models: $_\wedge_$, $\Diamond_$, and $\Box_$. Fill the table below as follows: in the cell corresponding to row (horizontal) x and column (vertical) y write YES if, according to the above grammar (without any additional operator precedence), the operators x and y can yield an ambiguity. Otherwise, write "——".

	$_\wedge_$	$\Diamond_$	$\Box_$
$_\wedge_$			
$\Diamond_$			
$\Box_$			

Solution:

	$_\wedge_$	$\Diamond_$	$\Box_$
$_\wedge_$	YES	YES	YES
$\Diamond_$	YES	-	-
$\Box_$	YES	-	-

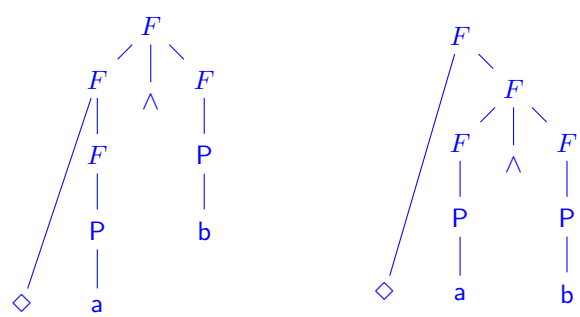
(c) If you have discovered at least one ambiguity in (b) that involves at least one modal operator ($\Box$ or $\Diamond$), showcase it by providing an example of a formula that has at least two distinct parse trees. Otherwise, explain in 2-3 lines why no formula exists that can have two parse trees.

Solution:

The ambiguity between $\wedge$ and $\Diamond$ is due to undefined precedence and can be showcased by the formula

$$\Diamond \mathsf{a} \wedge \mathsf{b}$$

and the two distinct parse trees:



The rest of the ambiguities can be similarly be showcased.

(d) If you have discovered at least one ambiguity in (b) that involves at least one modal operator ($\Box$ or $\Diamond$) change the grammar to remove the ambiguity. You do not need to solve all ambiguities you have discovered.

Explain in 2-3 lines the change that you have done and why it solves the ambiguity.

Solution:
To remove the illustrated ambiguity we could stratify the grammar to provide higher precedence to unary operators over conjunction:

$$
\begin{array}{rcll}
F_0 & \rightarrow & F_1 & \text{(Formulas with higher precedence)} \\
 & | & F_0 \wedge F_0 & \text{(... and ...)} \\
F_1 & \rightarrow & P & \text{(proposition...)} \\
 & | & \neg F_1 & \text{(not...)} \\
 & | & \Diamond F_1 & \text{(possibly...)} \\
 & | & \Box F_1 & \text{(necessarily ...)} \\
P & \rightarrow & \mathsf{a} \mid \mathsf{b} \mid \mathsf{c} & \text{(some atomic propositions)}
\end{array}
$$

Additional precedence ambiguities can be removed similarly. Associativity of $\wedge$ can be resolved by forbidding recursion on one side.

(e) Consider the following PDA which accepts by final state and is defined by the tuple $(\{q0, q_1, q_2\}, T, T \cup N, \delta, F, \{q_2\})$ where the transition function δ is defined as follows:

$$
\begin{array}{rcll}
\delta(q_0, u, F) & = & \{(q_0, F)\} & \text{if } u \in \{\Diamond, \Box\} \\
\delta(q_0, v, F) & = & \{(q_1, P)\} & \text{if } v \in \{\mathsf{a}, \mathsf{b}, \mathsf{c}\} \\
\delta(q_1, \epsilon, P) & = & \{(q_2, F)\} & \\
\delta(q_2, \wedge, F) & = & \{(q_0, F)\} &
\end{array}
$$

The PDA accepts the same language as the grammar. Check this by showing how the PDA can accept the string $\mathsf{a} \wedge \Diamond\Box\mathsf{b}$.

Solution:
We can provide the following run of the PDA

$$
\begin{array}{rll}
 & (q_0, & a \wedge \Diamond\neg\Box b, F) \\
\vdash & (q_1, & \wedge \Diamond\neg\Box b, P) \\
\vdash & (q_2, & \wedge \Diamond\neg\Box b, F) \\
\vdash & (q_0, & \Diamond\neg\Box b, F) \\
\vdash & (q_0, & \neg\Box b, F) \\
\vdash & (q_0, & \Box b, F) \\
\vdash & (q_0, & b, F) \\
\vdash & (q_1, & b, P) \\
\vdash & (q_2, & , F)
\end{array}
$$

(f) Can we build a DFA that recognizes the set of formulas accepted by the grammar described at the beginning of the exercise? Provide a short answer in 1-2 sentences. If your answer is positive, provide the DFA.

Solution: We can. The simplest solution is to notice that the PDA provided uses finite memory (the stack size is always 1), so necessarily one can find a DFA.

One way to obtain it (but this is not required as part of the solution) is by dropping the stack since it is uniquely determined by the current state. One obtains a simple ϵ-NFA from which we can easily remove the only ϵ transition.

Another way to look at it is to to observe that modal formulas can be seen as lists of $\wedge$-separated monadic formulas (i.e. formulas that contain atomic predicates preceded by any sequence of unary operators). We can express it with the regular expression

$$(\neg + \Diamond + \Box)^*(p + q + r)(\wedge(\neg + \Diamond + \Box)^*(p + q + r))^*$$

Exercise 7 (5%) Model Checking

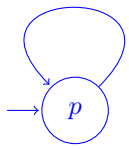
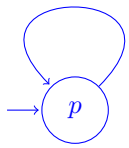
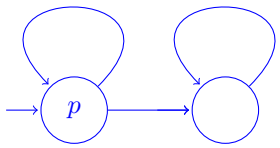
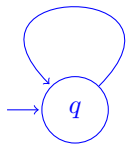
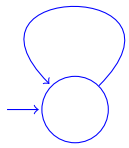
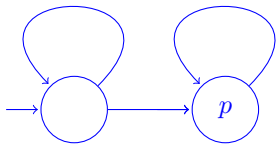
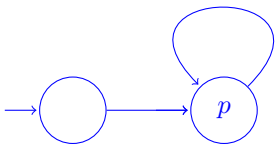
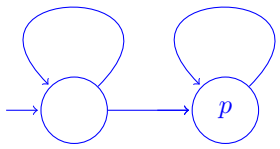
Fill the following table according to the rules explained below:

ϕ	$TS1 \models \phi$	$TS2 \not\models \phi$
$\mathsf{AF}p$		
$\mathsf{AG}p$		
$\mathsf{A(pUq)}$		
$\mathsf{AG(p \rightarrow AF\ q)}$		
$\neg\mathsf{p} \wedge \mathsf{AFp}$		

In column $TS1 \models \phi$ draw the smallest transition system $TS1$ that you can find that satisfies the formula ϕ. In column $TS2 \not\models \phi$ provide the mimimal extension you can do to TS1 (i.e. TS2 has the same states, transitions, initial states, labels, etc. but possibly more) that does *not* satisfy the formula. If you cannot find a transition system, write "——".

Here, "smallest" refers to the size of a transition system, counted as the number of states and transitions. As in the book, the transition systems cannot have stuck states.

Solution :

ϕ	$TS1 \models \phi$	$TS2 \not\models \phi$
AFp		———
AGp		
A(pUq)		———
AG(p → AF q)		
¬p ∧ AFp		

Exercise 8 (5%) Induction Proofs

Consider the function $\mathsf{dup}\colon \Sigma^* \to \Sigma^*$, given by the following recursive definition:

$$\mathsf{dup}(w) \;=\; \begin{cases} \varepsilon, & \text{if } w = \varepsilon \\ \mathsf{dup}(w')\ a\ a, & \text{if } w = w'a, a \in \Sigma, w' \in \Sigma^* \end{cases}$$

Furthermore, let $A = (Q, \Sigma, \delta, q_0, F)$ be a DFA. We transform A into another DFA A_2 as follows:

- $A_2 = (Q_2, \Sigma, \delta_2, q_0, F)$,
- $Q_2 = \{q_e\} \cup Q \cup \bigcup_{q \in Q} \{q_a \mid a \in \Sigma\}$ (where the error state q_e is fresh and the state q_a is fresh for every $q \in Q$ and $a \in \Sigma$),
- $\delta_2(q, a) = \begin{cases} q_a & \text{if } q \in Q \\ \delta(q', a) & \text{if } q \text{ is of the form } q'_a \text{ for some } q' \in Q \text{ and } a \in A \\ q_e & \text{otherwise} \end{cases}$

Complete the proof below to show that A_2 accepts $\mathsf{dup}(w)$ if and only if A accepts w by filling the blank lines.

We first show by induction on the structure of words that, for every word $w \in \Sigma^*$ and all states $q \in Q$ (but not necessarily the states in $Q_2 \setminus Q$), we have

$$\delta^*(q, w) = \delta_2^*(q, \mathsf{dup}(w)).$$

Induction base: For $w = \varepsilon$, we have **... (1-2 lines suffice)**

Induction hypothesis: For every word $w \in \Sigma^*$ and all states $q \in Q$,

$$\delta^*(q, w) = \delta_2^*(q, \mathsf{dup}(w)).$$

Induction step: Let $w = w'a$, where $a \in \Sigma$ and $w' \in \Sigma^*$,
... (5-6 lines suffice)

Hence, $\delta^*(q, w) = \delta_2^*(q, \mathsf{dup}(w))$, which finishes the proof by induction.

Now, using the above property for $q = q_0$, we conclude (for all words $w \in \Sigma^*$) that $\delta^*(q_0, w) \in F$ if and only if $\delta^*(q_0, \mathsf{dup}(w)) \in F$. Thus, A_2 accepts $\mathsf{dup}(w)$ if and only if A accepts w.

Solution:
Induction base: For $w = \varepsilon$, we have

$$\delta^*(q, \varepsilon) \;=\; q \;=\; \delta_2^*(q, \varepsilon) \;=\; \delta_2^*(q, \mathsf{dup}(\varepsilon)).$$

Induction hypothesis: For every word $w \in \Sigma^*$ and all states $q \in Q$,

$$\delta^*(q, w) = \delta_2^*(q, \mathsf{dup}(w)).$$

Induction step: Let $w = w'a$, where $a \in \Sigma$ and $w' \in \Sigma^*$.
Since A is a DFA, there is exactly one state q' such that $\delta^*(q, w'a) = q'$.
Analogously, there is exactly one state q'' such that $\delta^*(q, w') = q''$ and $\delta(q'', a) = q'$.
By I.H., this holds if and only if $\delta_2^*(q, \mathsf{dup}(w')) = q''$.
By construction of A_2, we also know $\delta_2(q'', a) = q''_a$ and $\delta_2(q''_a, a) = q'$.
Hence, $\delta^*(q, w'a) = q'$ if and only if $\delta_2^*(q, \mathsf{dup}(w'a)) = \delta_2^*(q, \mathsf{dup}(w')aa) = q'$
In particular, for $q = q_0$, we know (for all words $w \in \Sigma^*$) that $\delta^*(q_0, w) \in F$ if and only if $\delta^*(q_0, \mathsf{dup}(w)) \in F$.
Thus, A_2 accepts $\mathsf{dup}(w)$ if and only if A accepts w.